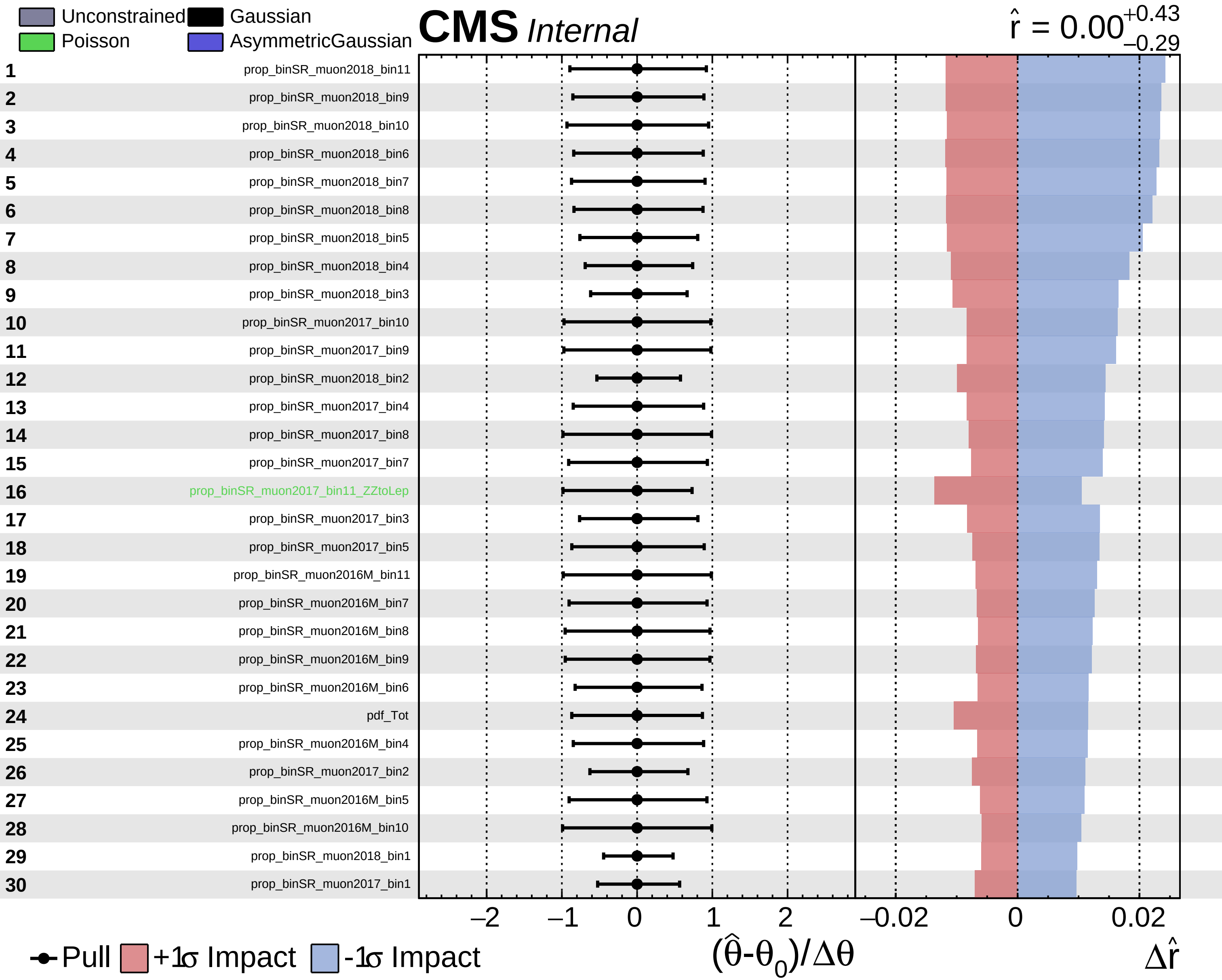
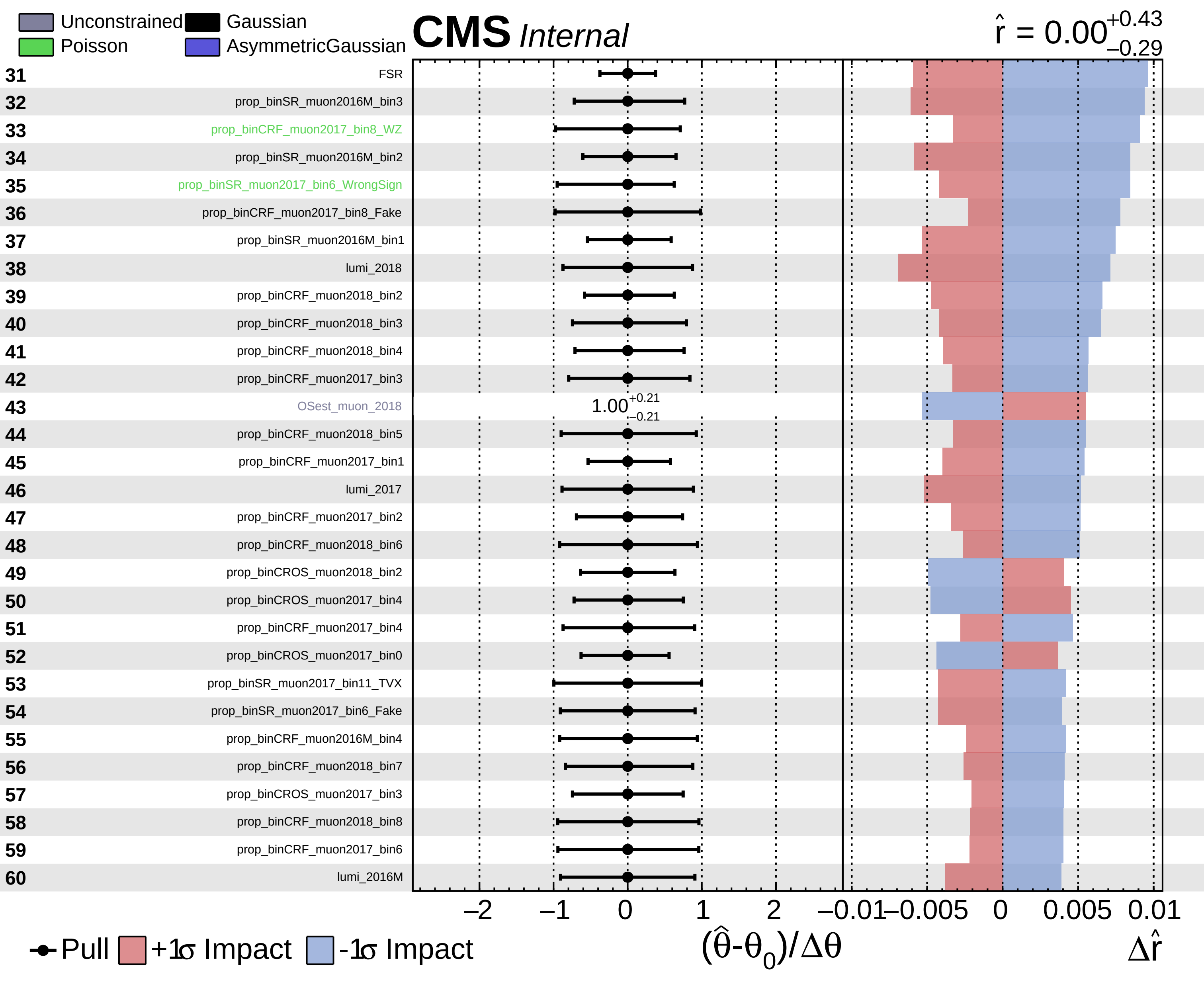
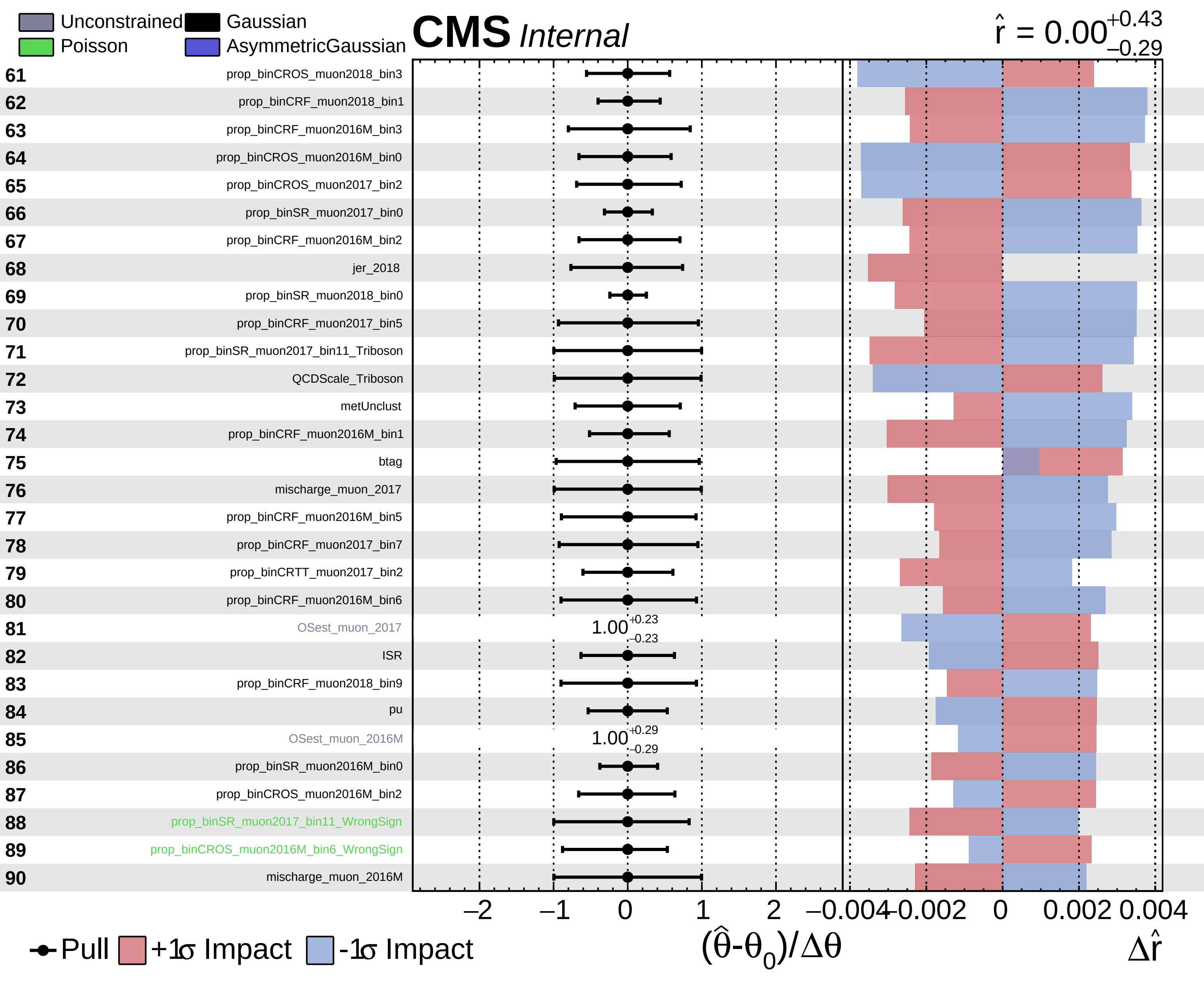
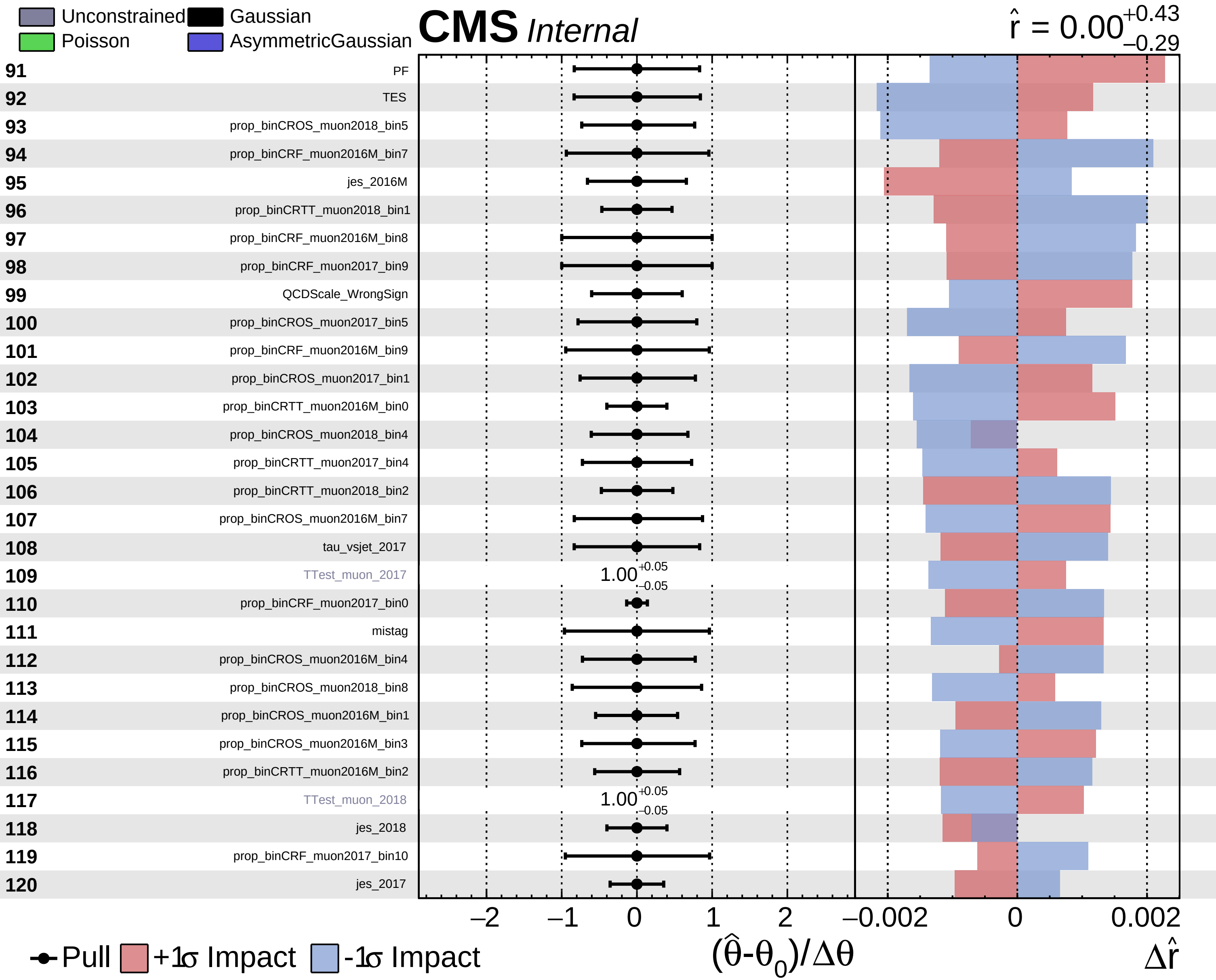


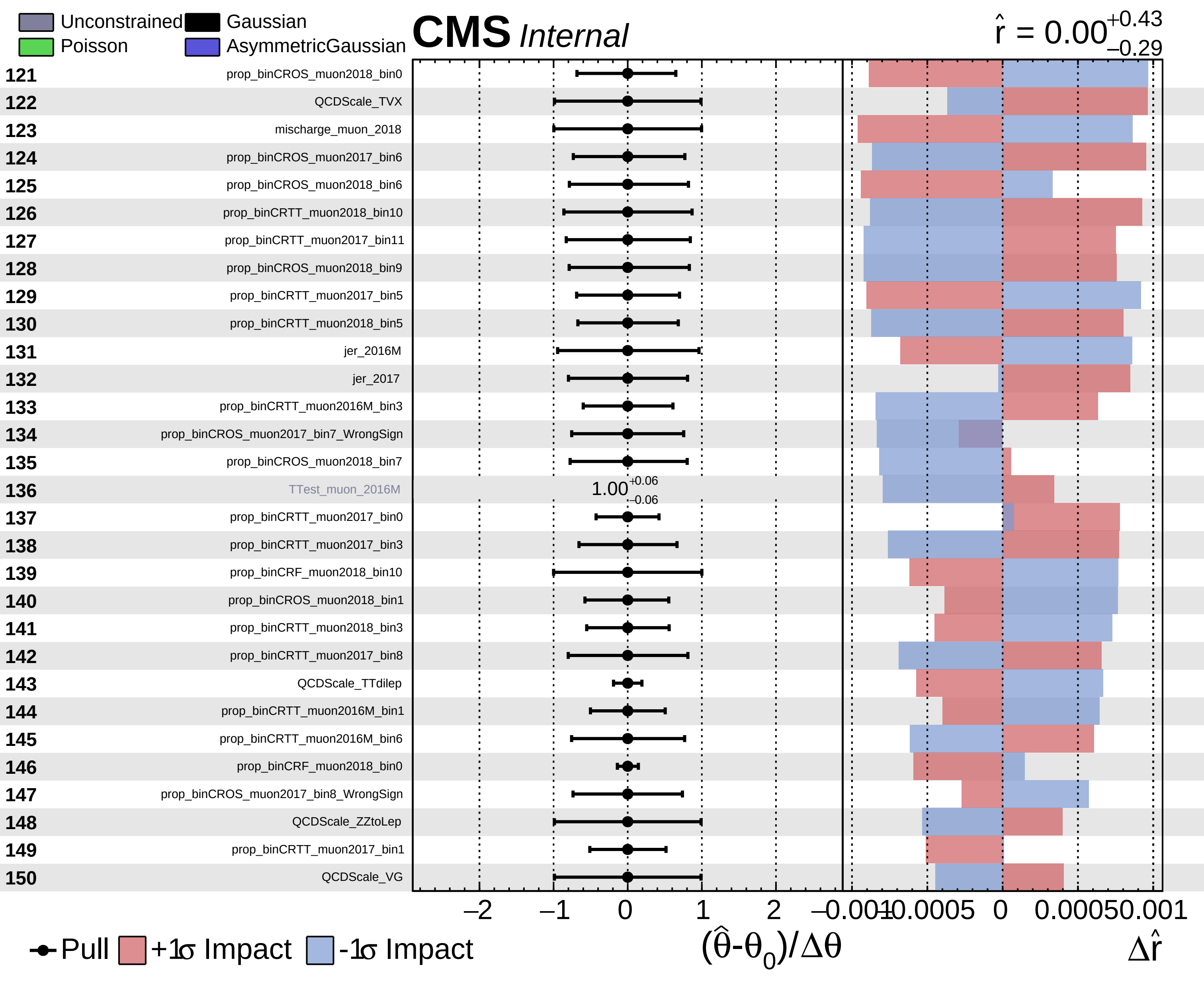
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.29}$
1 prop_binSR_muon2018_bin11
2 prop_binSR_muon2018_bin9
3 prop_binSR_muon2018_bin10
4 prop_binSR_muon2018_bin6
5 prop_binSR_muon2018_bin7
6 prop_binSR_muon2018_bin8
7 prop_binSR_muon2018_bin5
8 prop_binSR_muon2018_bin4
9 prop_binSR_muon2018_bin3
10 prop_binSR_muon2017_bin10
11 prop_binSR_muon2017_bin9
12 prop_binSR_muon2018_bin2
13 prop_binSR_muon2017_bin4
14 prop_binSR_muon2017_bin8
15 prop_binSR_muon2017_bin7
16 prop_binSR_muon2017_bin11_ZZtoLep
17 prop_binSR_muon2017_bin3
18 prop_binSR_muon2017_bin5
19 prop_binSR_muon2016M_bin11
20 prop_binSR_muon2016M_bin7
21 prop_binSR_muon2016M_bin8
22 prop_binSR_muon2016M_bin9
23 prop_binSR_muon2016M_bin6
24 pdf_Tot
25 prop_binSR_muon2016M_bin4
26 prop_binSR_muon2017_bin2
27 prop_binSR_muon2016M_bin5
28 prop_binSR_muon2016M_bin10
29 prop_binSR_muon2018_bin1
30 prop_binSR_muon2017_bin1
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

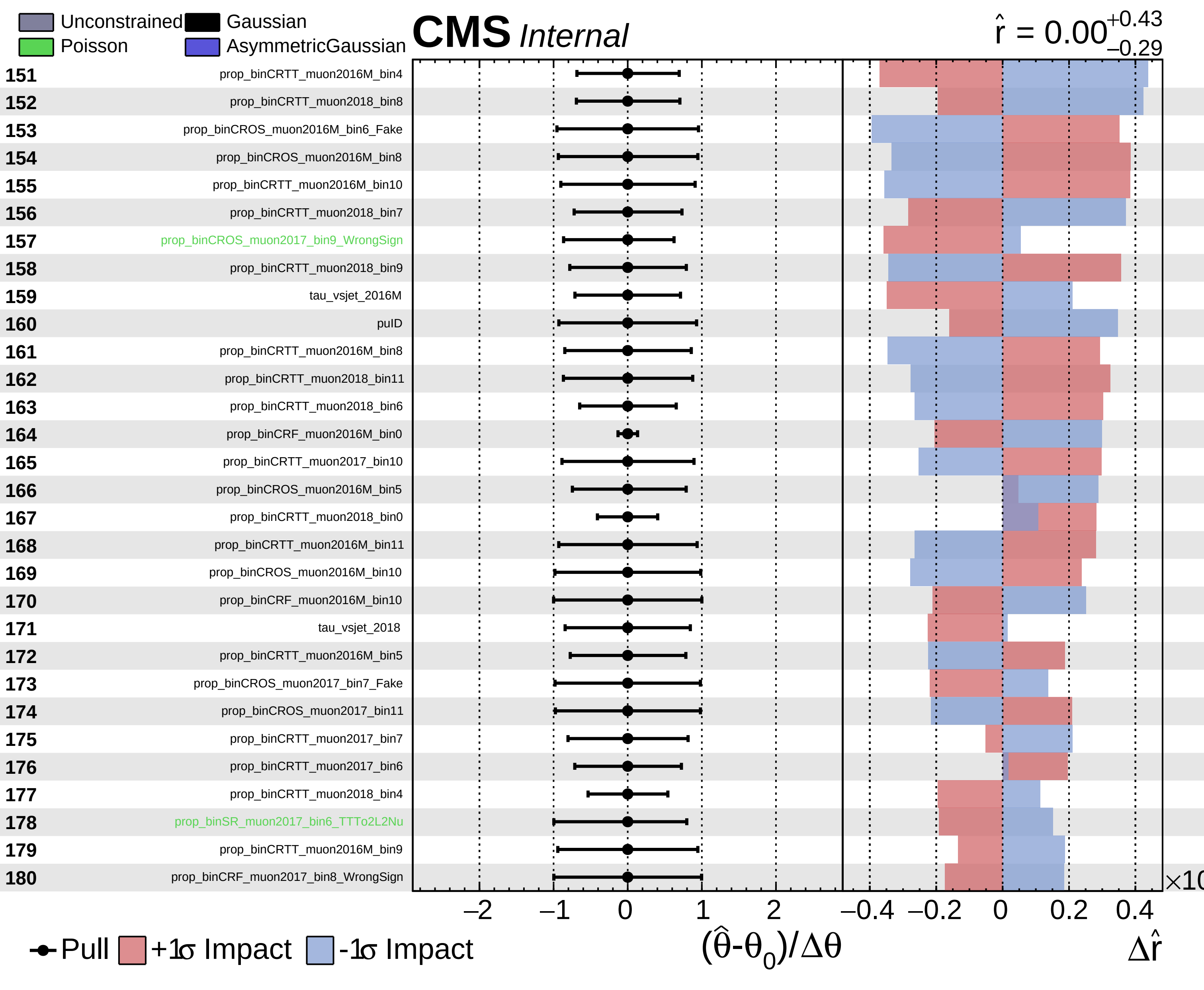

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.29}$
31 FSR
32 prop_binSR_muon2016M_bin3
33 prop_binCRF_muon2017_bin8_WZ
34 prop_binSR_muon2016M_bin2
35 prop_binSR_muon2017_bin6_WrongSign
36 prop_binCRF_muon2017_bin8_Fake
37 prop_binSR_muon2016M_bin1
38 lumi_2018
39 prop_binCRF_muon2018_bin2
40 prop_binCRF_muon2018_bin3
41 prop_binCRF_muon2018_bin4
42 prop_binCRF_muon2017_bin3
43 OSest_muon_2018
$1.00^{+0.21}_{-0.21}$
44 prop_binCRF_muon2018_bin5
45 prop_binCRF_muon2017_bin1
46 lumi_2017
47 prop_binCRF_muon2017_bin2
48 prop_binCRF_muon2018_bin6
49 prop_binCROS_muon2018_bin2
50 prop_binCROS_muon2017_bin4
51 prop_binCRF_muon2017_bin4
52 prop_binCROS_muon2017_bin0
53 prop_binSR_muon2017_bin11_TVX
54 prop_binSR_muon2017_bin6_Fake
55 prop_binCRF_muon2016M_bin4
56 prop_binCRF_muon2018_bin7
57 prop_binCROS_muon2017_bin3
58 prop_binCRF_muon2018_bin8
59 prop_binCRF_muon2017_bin6
60 lumi_2016M
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



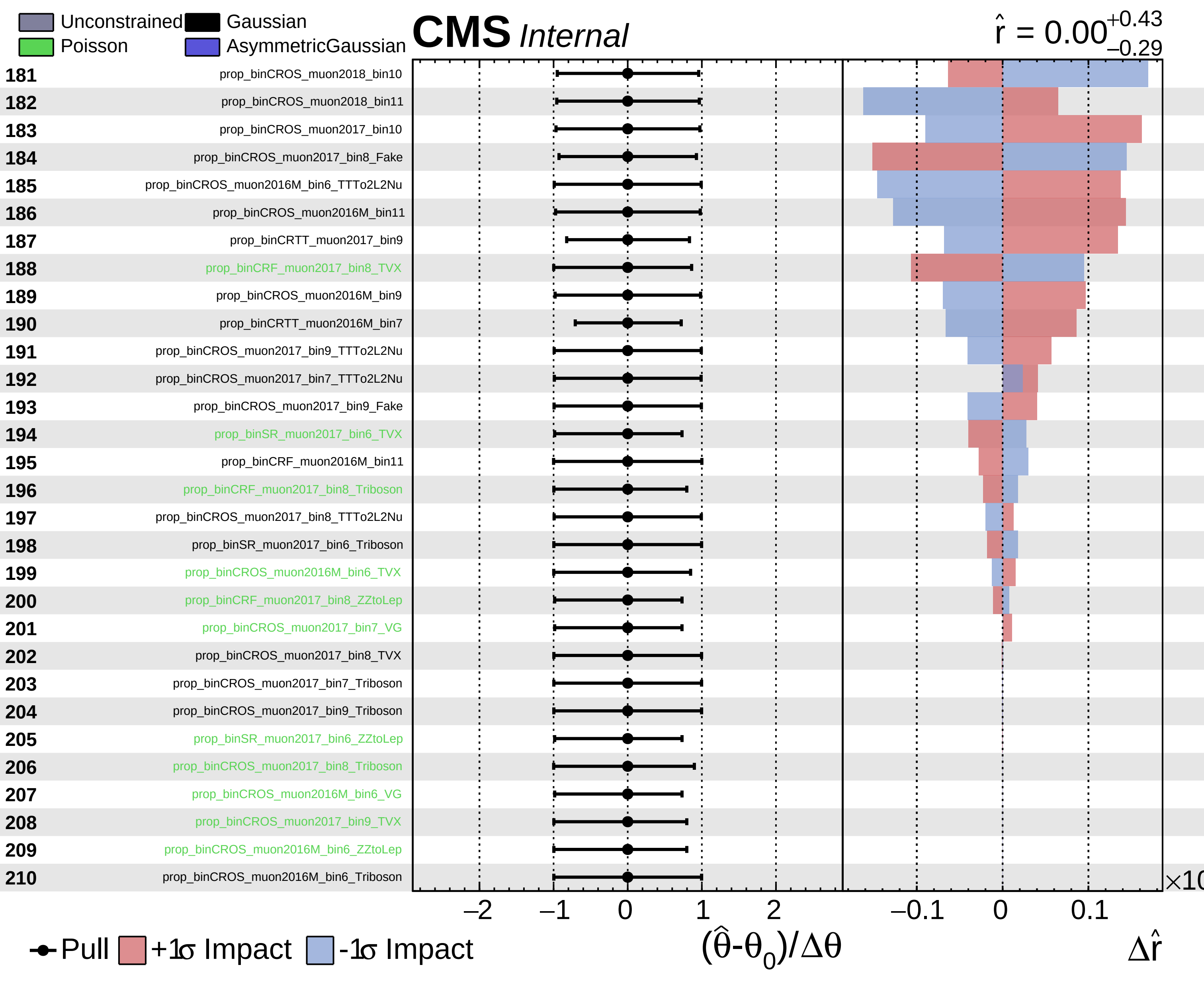
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.29}$
61 prop_binCROS_muon2018_bin3
62 prop_binCRF_muon2018_bin1
63 prop_binCRF_muon2016M_bin3
64 prop_binCROS_muon2016M_bin0
65 prop_binCROS_muon2017_bin2
66 prop_binSR_muon2017_bin0
67 prop_binCRF_muon2016M_bin2
68 jer_2018
69 prop_binSR_muon2018_bin0
70 prop_binCRF_muon2017_bin5
71 prop_binSR_muon2017_bin11_Triboson
72 QCDScale_Triboson
73 metUnclust
74 prop_binCRF_muon2016M_bin1
75 btag
76 mischarge_muon_2017
77 prop_binCRF_muon2016M_bin5
78 prop_binCRF_muon2017_bin7
79 prop_binCRTT_muon2017_bin2
80 prop_binCRF_muon2016M_bin6
81 OSest_muon_2017
$1.00^{+0.23}_{-0.23}$
82 ISR
83 prop_binCRF_muon2018_bin9
84 pu
85 OSest_muon_2016M
$1.00^{+0.29}_{-0.29}$
86 prop_binSR_muon2016M_bin0
87 prop_binCROS_muon2016M_bin2
88 prop_binSR_muon2017_bin11_WrongSign
89 prop_binCROS_muon2016M_bin6_WrongSign
90 mischarge_muon_2016M
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

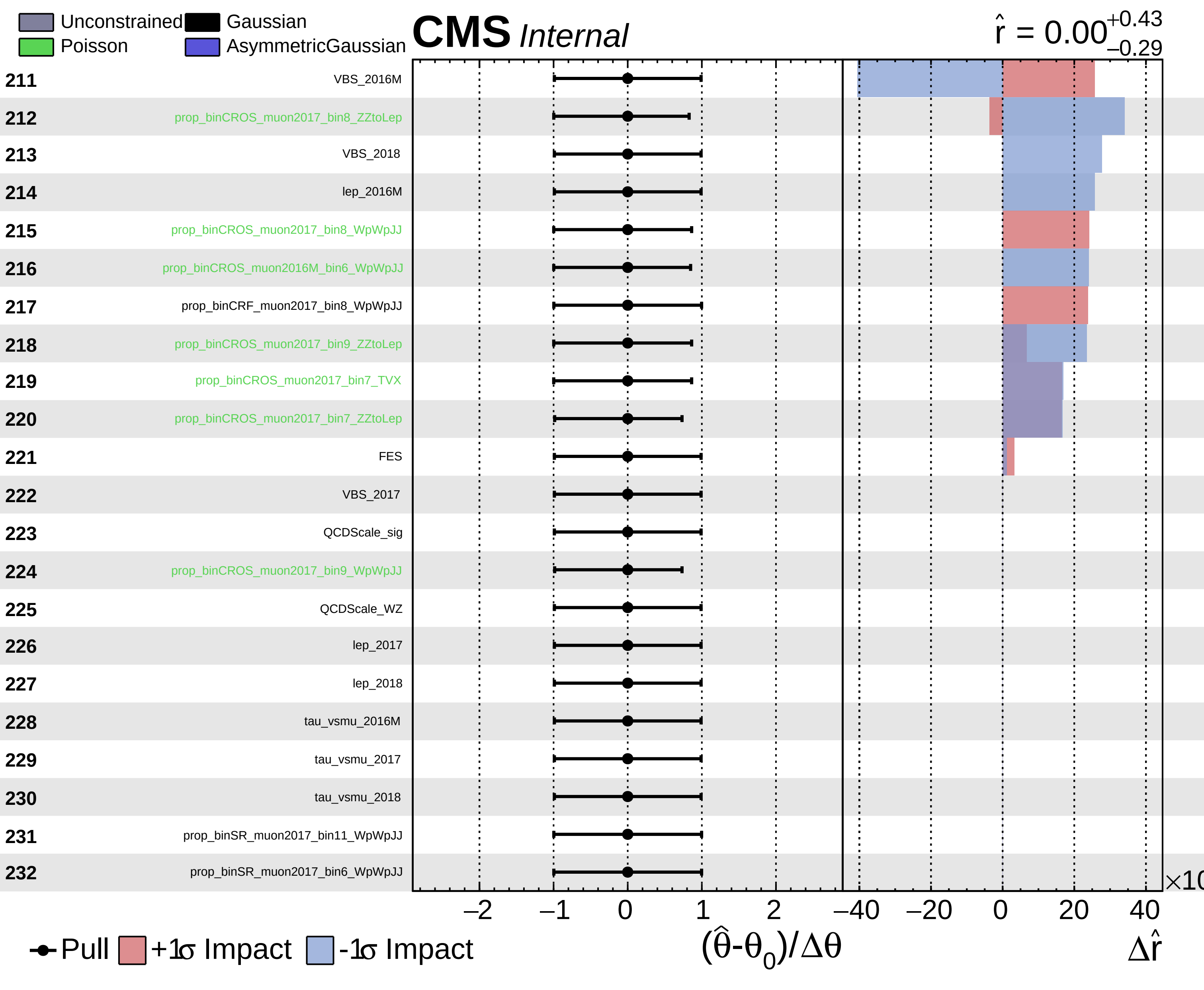

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.43 −0.29
91 PF
92 TES
93 prop_binCROS_muon2018_bin5
94 prop_binCRF_muon2016M_bin7
95 jes_2016M
96 prop_binCRTT_muon2018_bin1
97 prop_binCRF_muon2016M_bin8
98 prop_binCRF_muon2017_bin9
99 QCDScale_WrongSign
100 prop_binCROS_muon2017_bin5
101 prop_binCRF_muon2016M_bin9
102 prop_binCROS_muon2017_bin1
103 prop_binCRTT_muon2016M_bin0
104 prop_binCROS_muon2018_bin4
105 prop_binCRTT_muon2017_bin4
106 prop_binCRTT_muon2018_bin2
107 prop_binCROS_muon2016M_bin7
108 tau_vsjet_2017
109 TTest_muon_2017
1.00 +0.05 −0.05
110 prop_binCRF_muon2017_bin0
111 mistag
112 prop_binCROS_muon2016M_bin4
113 prop_binCROS_muon2018_bin8
114 prop_binCROS_muon2016M_bin1
115 prop_binCROS_muon2016M_bin3
116 prop_binCRTT_muon2016M_bin2
117 TTest_muon_2018
1.00 +0.05 −0.05
118 jes_2018
119 prop_binCRF_muon2017_bin10
120 jes_2017
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.29}$
121 prop_binCROS_muon2018_bin0
122 QCDScale_TVX
123 mischarge_muon_2018
124 prop_binCROS_muon2017_bin6
125 prop_binCROS_muon2018_bin6
126 prop_binCRTT_muon2018_bin10
127 prop_binCRTT_muon2017_bin11
128 prop_binCROS_muon2018_bin9
129 prop_binCRTT_muon2017_bin5
130 prop_binCRTT_muon2018_bin5
131 jer_2016M
132 jer_2017
133 prop_binCRTT_muon2016M_bin3
134 prop_binCROS_muon2017_bin7_WrongSign
135 prop_binCROS_muon2018_bin7
136 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
137 prop_binCRTT_muon2017_bin0
138 prop_binCRTT_muon2017_bin3
139 prop_binCRF_muon2018_bin10
140 prop_binCROS_muon2018_bin1
141 prop_binCRTT_muon2018_bin3
142 prop_binCRTT_muon2017_bin8
143 QCDScale_TTdilep
144 prop_binCRTT_muon2016M_bin1
145 prop_binCRTT_muon2016M_bin6
146 prop_binCRF_muon2018_bin0
147 prop_binCROS_muon2017_bin8_WrongSign
148 QCDScale_ZZtoLep
149 prop_binCRTT_muon2017_bin1
150 QCDScale_VG
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.43 −0.29
151 prop_binCRTT_muon2016M_bin4
152 prop_binCRTT_muon2018_bin8
153 prop_binCROS_muon2016M_bin6_Fake
154 prop_binCROS_muon2016M_bin8
155 prop_binCRTT_muon2016M_bin10
156 prop_binCRTT_muon2018_bin7
157 prop_binCROS_muon2017_bin9_WrongSign
158 prop_binCRTT_muon2018_bin9
159 tau_vsjet_2016M
160 puID
161 prop_binCRTT_muon2016M_bin8
162 prop_binCRTT_muon2018_bin11
163 prop_binCRTT_muon2018_bin6
164 prop_binCRF_muon2016M_bin0
165 prop_binCRTT_muon2017_bin10
166 prop_binCROS_muon2016M_bin5
167 prop_binCRTT_muon2018_bin0
168 prop_binCRTT_muon2016M_bin11
169 prop_binCROS_muon2016M_bin10
170 prop_binCRF_muon2016M_bin10
171 tau_vsjet_2018
172 prop_binCRTT_muon2016M_bin5
173 prop_binCROS_muon2017_bin7_Fake
174 prop_binCROS_muon2017_bin11
175 prop_binCRTT_muon2017_bin7
176 prop_binCRTT_muon2017_bin6
177 prop_binCRTT_muon2018_bin4
178 prop_binSR_muon2017_bin6_TTTo2L2Nu
179 prop_binCRTT_muon2016M_bin9
180 prop_binCRF_muon2017_bin8_WrongSign
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.29}$
181 prop_binCROS_muon2018_bin10
182 prop_binCROS_muon2018_bin11
183 prop_binCROS_muon2017_bin10
184 prop_binCROS_muon2017_bin8_Fake
185 prop_binCROS_muon2016M_bin6_TTTo2L2Nu
186 prop_binCROS_muon2016M_bin11
187 prop_binCRTT_muon2017_bin9
188 prop_binCRF_muon2017_bin8_TVX
189 prop_binCROS_muon2016M_bin9
190 prop_binCRTT_muon2016M_bin7
191 prop_binCROS_muon2017_bin9_TTTo2L2Nu
192 prop_binCROS_muon2017_bin7_TTTo2L2Nu
193 prop_binCROS_muon2017_bin9_Fake
194 prop_binSR_muon2017_bin6_TVX
195 prop_binCRF_muon2016M_bin11
196 prop_binCRF_muon2017_bin8_Triboson
197 prop_binCROS_muon2017_bin8_TTTo2L2Nu
198 prop_binSR_muon2017_bin6_Triboson
199 prop_binCROS_muon2016M_bin6_TVX
200 prop_binCRF_muon2017_bin8_ZZtoLep
201 prop_binCROS_muon2017_bin7_VG
202 prop_binCROS_muon2017_bin8_TVX
203 prop_binCROS_muon2017_bin7_Triboson
204 prop_binCROS_muon2017_bin9_Triboson
205 prop_binSR_muon2017_bin6_ZZtoLep
206 prop_binCROS_muon2017_bin8_Triboson
207 prop_binCROS_muon2016M_bin6_VG
208 prop_binCROS_muon2017_bin9_TVX
209 prop_binCROS_muon2016M_bin6_ZZtoLep
210 prop_binCROS_muon2016M_bin6_Triboson
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.29}$
211 VBS_2016M
212 prop_binCROS_muon2017_bin8_ZZtoLep
213 VBS_2018
214 lep_2016M
215 prop_binCROS_muon2017_bin8_WpWpJJ
216 prop_binCROS_muon2016M_bin6_WpWpJJ
217 prop_binCRF_muon2017_bin8_WpWpJJ
218 prop_binCROS_muon2017_bin9_ZZtoLep
219 prop_binCROS_muon2017_bin7_TVX
220 prop_binCROS_muon2017_bin7_ZZtoLep
221 FES
222 VBS_2017
223 QCDScale_sig
224 prop_binCROS_muon2017_bin9_WpWpJJ
225 QCDScale_WZ
226 lep_2017
227 lep_2018
228 tau_vsmu_2016M
229 tau_vsmu_2017
230 tau_vsmu_2018
231 prop_binSR_muon2017_bin11_WpWpJJ
232 prop_binSR_muon2017_bin6_WpWpJJ
−2
−1
0
1
2
−40
−20
0
20
40
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$